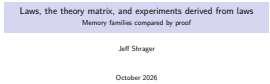
Laws, the theory matrix, and experiments derived from laws

Memory families compared by proof

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This talk is about comparing families of memory models by proof. I'll describe an abstract memory, six laws a memory may have, and four families of models, then the matrix of which family has which law, how the laws were tested, and what the evidence about people says. Then I'll turn to experiments: one study-test experiment run with four memories, and a counterbalanced experiment derived from two laws that disagree, with a response model that connects it to what participants report.

A memory, abstractly

- **Study** an item: the memory's state changes.
- **Retrieve** with a probe: a probability distribution over stored items, or nothing.

Items are binary feature codes; similarity is the number of mismatching features.

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Start with what a memory is, abstractly. You can study an item, which changes the memory's state, and you can retrieve with a probe, which gives a probability distribution over stored items, or nothing. Items are binary feature codes, and similarity is the number of features on which two codes differ. Every law I'll show is a property of this interface. None of them names a mechanism, but they do assume this representation, and that's a limitation I'll come back to.

Six laws

Certainty	An item just studied is retrieved by itself with probability 1.
Stability	Studying z leaves the chance that y retrieves itself unchanged.
Order-freedom	Swapping the last two study events leaves retrieval unchanged, for every probe.
Recency	Of two new items studied last, a probe equally similar to both retrieves the later more often.
Stored-only	Nothing is retrieved that was never studied.
Determinacy	Every retrieval has a single outcome.

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To compare memories in general, we need a vocabulary that doesn't mention any mechanism. These are laws: small properties a memory may or may not have. Certainty: something just studied is retrieved for certain. Stability: studying one thing doesn't change another's retrieval. Order-freedom; recency; stored-only, meaning nothing unstudied is ever retrieved; and determinacy, a single outcome. Predictions are then proved from laws, not from particular models.

Four families

- **Discrimination** (EPAM): a net of feature tests; learning grows a test where codes are confused.
- **Activation** (ACT-R declarative memory): each use adds activation that decays; partial matching; a failure weight. Decay 1, temperature 1: exactly rational.
- **Strength with a criterion** (after DOA): strengths only add; sample by strength, discounted by mismatch; state if above criterion.
- **Global matching** (MINERVA 2): every study stores a trace; a probe activates all traces by the cube of similarity; memory returns the *echo*, their weighted blend.

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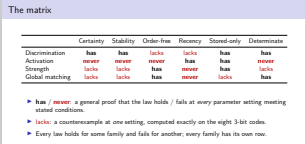
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And instead of individual models, we formalize families of models, each once, with its parameters free. Discrimination nets, like EPAM. Activation memory, like ACT-R's. Strength memory with a criterion, after the associations model. And global matching, MINERVA 2, where every study leaves a trace and a probe returns a blend of all of them. A published model is then a choice of parameters within its family.

The matrix

	Certainty	Stability	Order-free	Recency	Stored-only	Determinate
Discrimination	**has**	**has**	lacks	lacks	**has**	**has**
Activation	**never**	**never**	**never**	**has**	**has**	**never**
Strength	lacks	lacks	**has**	**never**	**has**	lacks
Global matching	lacks	lacks	**has**	**never**	lacks	**has**

- **has** / **never**: a general proof that the law holds / fails at *every* parameter setting meeting stated conditions.
- lacks: a counterexample at *one* setting, computed exactly on the eight 3-bit codes.
- Every law holds for some family and fails for another; every family has its own row.

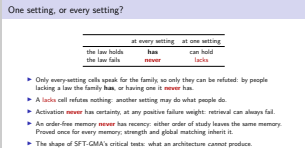


Put them together and you get a matrix. Every cell is a theorem, and there are three kinds. Has, in black, means a general proof that the family has the law at every parameter setting that meets stated conditions. Never, in bold red, means a general proof that it fails at every such setting. Lacks, in plain red, means a counterexample at one setting. ACT-R, for instance, is never certain, never stable, never order-free and never determinate, at any positive failure weight; the other families' lacks are single counterexamples. Every law separates some families, and every family has its own row. I'll be candid: some of these laws are stated too narrowly, all-or-none where they should be graded, and fixing that is the next piece of work. But it's the structure I want you to see.

One setting, or every setting?

	at every setting	at one setting
the law holds	**has**	can hold
the law fails	**never**	lacks

- Only every-setting cells speak for the family, so only they can be refuted: by people lacking a law the family **has**, or having one it **never** has.
- A lacks cell refutes nothing: another setting may do what people do.
- Activation **never** has certainty, at any positive failure weight: retrieval can always fail.
- An order-free memory **never** has recency: either order of study leaves the same memory. Proved once for every memory; strength and global matching inherit it.
- The shape of SFT-GMA's critical tests: what an architecture *cannot* produce.



Why distinguish never from lacks? Because they say different things about a family. A counterexample at one setting shows the family doesn't guarantee the law, but another setting might have it. Only a claim about every setting speaks for the whole family, so only has and never can be refuted. If people lack a law, a family that has it everywhere is in trouble; if people have it, a family that never has it is. Two of the never cells are worth a word. ACT-R is never certain simply because retrieval can always fail. And any order-free memory never has recency, for a general reason: studying two items in either order leaves the same memory, so neither order can favour the later item. That one is proved once, for every memory, and strength and global matching inherit it. This is the same shape as the critical tests of Fisher and colleagues: what an architecture cannot produce, under any auxiliary choice.

The laws were tested, and failed once

- The first four laws were chosen with three families in view.
- Global matching was added as a test. On those four laws, it had *exactly* the row of strength memory.
- So the laws missed something. What is distinctive about global matching? Its output is a blend.
- Study 110, 101, 011; probe with 111 (their prototype, never studied): MINERVA returns 111.
- Hence the stored-only law, which separates them.

Rule: a law enters only with a witness on each side.

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We also test the laws themselves. The first four were chosen with three families in mind. When we added global matching as a test, it looked exactly like strength memory on all four. So the laws were missing something, and what's distinctive about global matching is that it returns a blend. Study three variants of a pattern, probe with their prototype, and MINERVA returns the prototype, which it never saw. That forced the stored-only law. The rule now is that a law enters the library only with a family on each side.

Against the evidence

	Certainty	Stability	Order-free	Recency	Stored-only	Determinate
People (judged)	*lacks*	*mixed*	*lacks*	*has*	*mixed*	*mixed*

- A judgment, not a theorem, with sources, kept apart from the proofs.
- Each entry states its **bridge**: how a measure (a report) is taken to show a law (about retrieval).

Family	refuted on (an every-setting cell)	open on (one setting only)
Discrimination	certainty	recency
Activation	—	—
Strength	order-freedom, recency	—
Global matching	order-freedom, recency	—

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So the matrix carries a row for people: a judgment, with sources, kept apart from the theorems. Each entry states its bridge, that is, how a measure, which is always a report, is taken to show something about retrieval. Endorsing an unseen prototype as old, for example, doesn't show you retrieved it. Against the settled entries, only the every-setting cells can refute. EPAM is refuted on certainty: it has it everywhere, and people lack it. Its recency conflict is only open, since that cell is one counterexample. Strength and global matching are each refuted twice: they're order-free everywhere, which people aren't, and they never have recency, which people do. ACT-R fits every settled entry. All of these are hypotheses until the bridges are formalized.

One experiment, four memories: study, then test

Study 110, 101, 011. Probe each; then the lures 111 (prototype) and 000.

Memory	curve by position	111 called old	111 retrieves itself	000 called old
EPAM	$1, 1, 1$	1	0	1
ACT-R	$\frac{40}{97}, \frac{15}{28}, \frac{120}{157}$	$\frac{55}{61}$	0	$\frac{55}{67}$
Strength	$\frac{8}{17}, \frac{8}{17}, \frac{8}{17}$	1	0	1
MINERVA	$1, 1, 1$	0	1	0

- Derived from laws: EPAM's flat curve (certainty + stability $\Rightarrow$ flat at 1, for any memory).
- Computed: everything else. Recency does not imply a rising self-retrieval curve.
- "Called old" is a report; stored-only is about retrieval. EPAM calls the prototype old by retrieving a studied item.

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EPAM	1, 1, 1	1	0	1
ACT-R	$\frac{\ln 3}{\ln 4}$, $\frac{\ln 3}{\ln 4}$, $\frac{\ln 3}{\ln 4}$	$\frac{\ln 3}{\ln 4}$	0	$\frac{\ln 3}{\ln 4}$
Strength	$\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{4}$	1	0	1
MINERVA	1, 1, 1	0	1	0

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Here is a study-test experiment written once and run with all four memories. Study three items, then probe with each, and with two lures: their prototype, which was never studied, and an item unlike all three. Only one entry here is derived from laws: EPAM's flat curve, because certainty and stability imply a flat curve at one for any memory. Everything else is computed from the mechanisms. And notice the bridge problem: EPAM and strength memory call the unseen prototype old with certainty, by retrieving a studied item, while MINERVA, the one family that retrieves the prototype itself, doesn't call it old under this crude report rule.

Reversing the workflow

Usually: choose an experiment, compute each theory's predictions.

Instead:

1. find two laws that disagree;
2. derive a design on which they make incompatible predictions;
3. prove the separation, for every memory with either law;
4. state the response model that connects it to what participants say.

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Laws let us reverse the usual workflow. Instead of choosing an experiment and computing what each model predicts, we start from two laws that disagree, derive a design on which they make incompatible predictions, prove the separation for every memory with either law, and then state the response model that connects it to what participants actually say.

Recency against order-freedom

Design. Two new items a, b; a probe x equally similar to both. Condition AB studies a then b; condition BA studies b then a; both probe with x.

$$D = P(\text{retrieve } b) - P(\text{retrieve } a), \qquad \Delta = D_{AB} - D_{BA}.$$

Theorems, for any memory:

- order-freedom $\Rightarrow \Delta = 0$;
- recency $\Rightarrow \Delta > 0$;
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Recency and order-freedom disagree. Take two new items, A and B, and a probe equally similar to both. Study A then B, or B then A, and then probe. Delta measures how much more B is favoured after AB than after BA. From the laws alone: order-freedom gives a delta of zero, and recency gives a delta greater than zero, so no memory can have both laws. These proofs never mention EPAM, or ACT-R, or any parameter.

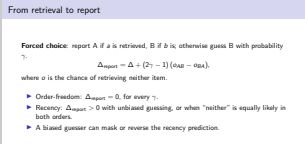
From retrieval to report

Forced choice: report A if a is retrieved, B if b is; otherwise guess B with probability γ.

$$\Delta_{\text{report}} = \Delta + (2\gamma - 1)\,(o_{AB} - o_{BA}),$$

where o is the chance of retrieving neither item.

- Order-freedom: $\Delta_{\text{report}} = 0$, for every γ.
- Recency: $\Delta_{\text{report}} > 0$ with unbiased guessing, or when "neither" is equally likely in both orders.
- A biased guesser can mask or reverse the recency prediction.



But participants give reports, not retrievals. So we add a response model: forced choice, guessing B with probability gamma when neither item comes to mind. Then the reported effect is delta plus a bias term. Order-freedom still predicts zero, whatever the bias. But recency's prediction survives only if guessing is unbiased, or if neither item is equally likely to come to mind in both orders. The bridge from theory to data comes with its own condition, and the theorem states it.

The same design, four memories

Memory	Δ, no prior item	Δ, after 000	why
ACT-R	$10/17$		recency, so $\Delta > 0$
Strength	0	0	order-free
MINERVA	0	0	order-free
EPAM	0	-2	neither law: computed

- ACT-R's reported effect is $10/17$ for every guessing bias: both orders leave the same chance of "neither".
- EPAM shows the most extreme *primacy* possible. After 000, whichever item is learned first is discriminated from it, and the probe always retrieves that one. Its order effects come from the shape of the net, not from time.

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Here is the same design run with four memories. ACT-R shows recency, ten seventeenths, and that holds for any guessing bias. Strength memory and MINERVA show no effect, as their order-freedom requires. EPAM has neither law, and it surprises us: with one prior item, it shows the most extreme primacy possible. Whichever item was learned first always wins, because of the shape of the net, not anything about time.

Lessons, and next

- The durable results are conditional: laws $\Rightarrow$ predictions, families $\Rightarrow$ laws.
- The empirical row will change; when it does, the matrix says what is refuted.
- **Observational semantics** is the main gap: decision and report models, so retrieval laws predict responses.
- **Parameter regions** of the "lacks" cells; **graded laws**; **more families** as tests (REM, connectionist memory, ACT-R with blending).
- **Beyond memory**: strategy choice, attention, production systems.

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To sum up. The durable results are conditional: laws imply predictions, and families have or lack laws. The empirical row will change, and when it does, the matrix says what is refuted. The main gap is what we might call observational semantics: decision and report models, so that laws about retrieval predict responses. Beyond that: the parameter regions of the lacks cells, graded laws in place of all-or-none ones, more families as tests, such as REM, a connectionist memory and ACT-R with blending, and other domains, such as strategy choice, attention and production systems.